Ghazal Khalighinejad

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Education

Duke University September 2021 - December 2026 *(Expected)*
Ph.D. Student in Computer Science; GPA: 3.86/4
Advisors: Bhuwan Dhingra, Sam Wiseman

Sharif University of Technology September 2017 - May 2021
Bachelor of Science in Computer Science

Research Interest

My research focuses on Natural Language Processing and Multimodal Large Language Models, with an emphasis on improving embedding methods for multimodal long-document retrieval and building LLM agents for scientific discovery. In the past, I've worked on structured information extraction from scientific articles, benchmarking the reasoning of multimodal models, and the algorithmic capabilities of transformers.

Positions

Simons Foundation & Polymathic AI *Junior Research Scientist* May 2026 – August 2026
Host: Shirley Ho; collaborators: Irina Espejo, Haewon Jeong
Built a closed-loop system in which an LLM coding agent iteratively proposes, implements, and evaluates new objective functions and search algorithms for recovering molecular structures from NMR spectra, keeping only changes that improve a held-out metric. Improved structure-recovery rate from 54% to 81% on a held-out validation set over the baseline.

Google Cloud *Student Researcher* May 2025 – November 2025
Host: Hannah Lin; collaborators: Milad Hashemi, Arman Hasanzadeh
Developed a paper-to-code retrieval system by fine-tuning a code-embedding model. Created a benchmark for evaluating ML concept–to–code mapping.

Adobe Research *Research Intern* May 2024 – August 2024
Host: Somdeb Sarkhel
Built a text-to-video retrieval system that identifies relevant shots and assembles them into coherent sequences using a dynamic programming algorithm to select and order shots.

Selected Publications

Full list on Google Scholar.

1. Document-as-Image Representations Fall Short for Scientific Retrieval. Ghazal Khalighinejad, Raghuveer Thirukovalluru, Alexander H. Oh, Bhuwan Dhingra. ***COLM* 2026.**
2. MatViX: Multimodal Information Extraction from Visually Rich Articles. Ghazal Khalighinejad, Sharon Scott, Ollie Liu, Kelly L. Anderson, Rickard Stureborg, Aman Tyagi, Bhuwan Dhingra. ***NAACL* 2025,** Oral Presentation.
3. Training Neural Networks as Recognizers of Formal Languages. Alexandra Butoi, Ghazal Khalighinejad, Anej Svete, Josef Valvoda, Ryan Cotterell, Brian DuSell. ***ICLR* 2025.**
4. IsoBench: Benchmarking Multimodal Foundation Models on Isomorphic Representations. Deqing Fu*, Ruohao Guo*, Ghazal Khalighinejad*, Ollie Liu*, Bhuwan Dhingra, Dani Yogatama, Robin Jia, Willie Neiswanger (*Equal contribution). ***COLM* 2024.**
5. Extracting Polymer Nanocomposite Samples from Full-Length Documents. Ghazal Khalighinejad, Defne Circi, L.C. Brinson, Bhuwan Dhingra. ***Findings of ACL* 2024**.
6. Approximating CKY with Transformers. Ghazal Khalighinejad, Ollie Liu, Sam Wiseman. ***Findings of EMNLP* 2023**.

Workshop & Journal Papers

1. It's LIT! LLMs with Interpretable Tool Calling. Ruixin Zhang, Jon Donnelly, Zhicheng Guo, Ghazal Khalighinejad, Haiyang Huang, Alina Jade Barnett, Cynthia Rudin. ***MTI-LLM @ NeurIPS* 2025**.
2. How Well Do Large Language Models Understand Tables in Materials Science? Defne Circi, Ghazal Khalighinejad, Anlan Chen, Bhuwan Dhingra, L.C. Brinson. *Integrating Materials and Manufacturing Innovation (**IMMI 2024**)*.
3. Exploring the Effect of Frequency Resolution in FNet. Gregory Szumel, Ghazal Khalighinejad, Rickard Stureborg and Sam Wiseman. ***SustaiNLP @ ACL* 2023**.
4. Retrieval of Synthesis Parameters of Polymer Nanocomposites using LLMs. Defne Circi, Ghazal Khalighinejad, Shruti Badhwar, Bhuwan Dhingra, L. Brinson. ***AI4MAT @ NeurIPS* 2023**.
5. Galloping in FastGrowth Natural Merge Sorts. Elahe Ghasemi, Vincent Jugé, Ghazal Khalighinejad, Helia Yazdanyar. ***Algorithmica* 2024** and ***ICALP* 2022**.

Research Experience

Multimodal Retrieval and Information Extraction 2024-2026

- Built a benchmark and hierarchical evaluation suite for extracting structured data from visually rich scientific articles, enabling end-to-end assessment of vision-language models on full-length documents.
- Benchmarked text-only, image-based, and interleaved representations across single- and multi-vector retrievers, showing document-as-image embeddings degrade on text-rich scientific papers while interleaved text+image inputs outperform them with no additional training.

Algorithmic Reasoning in Transformers 2022-2023

- Trained transformers to approximate CKY parsing, replacing CKY in modern constituency parsers without accuracy loss and improving runtime from cubic to quadratic dependence on sentence length.

Awards & Achievements

Oral Presentation NAACL 2025

LLMs for Materials Hackathon, Anthropic Award Won Anthropic API credits.

aiM National Science Foundation Fellow Full-tuition scholarship and funding for research in AI + Materials.

ACM-W Research Conference Scholarship Travel scholarship to attend NeurIPS 2022.

CRA-WP Scholarship Travel scholarship to attend CRA-WP Grad Cohort for Women Workshop.

Teaching Assistance Experience

Natural Language Processing (Graduate Course), Duke University, Fall 2022
Algorithms (Graduate Course), Duke University, Spring 2021
Game Theory (Graduate Course), Sharif University of Technology, Spring 2020
Advanced Programming (Undergraduate Course), Sharif University of Technology, Fall 2019

Doctoral Courseworks

Neurosymbolic Machine Learning, Natural Language Processing, Deep Learning, Causality and Interpretability, Probability and Statistics, Algorithms

Services

Reviewer COLM 2026, CVPR 2026, COLM 2025, ACL Rolling Review 2024, 2025

Skills

Programming: Python, Java

Libraries: PyTorch, JAX, TensorFlow